Linear Algebra - Written exam - 18 June 2026 - Duration: 120 minutes - Total 30 points

Exercise 1 (6 pts). Compute the determinant of A = [[2,1,0],[1,3,1],[0,1,2]] and say whether A is invertible.

Exercise 2 (8 pts). Find eigenvalues and eigenvectors of B = [[4,1],[2,3]]. Is B diagonalizable? Justify.

Exercise 3 (8 pts). Let W = span{(1,1,0),(0,1,1)}. Find an orthonormal basis of W with Gram-Schmidt and the orthogonal projection of v=(1,2,3) onto W.

Exercise 4 (8 pts). For which values of k does the system x+y+kz=1, x+ky+z=1, kx+y+z=1 have unique, infinite or no solutions?